Moad Essabbar

Ph.D. candidate in electronics, components and systems
Embedded systems and industrial IT engineer

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Experience

- **Ph.D. candidate in electronics, components and systems** — Montpellier
  Institstut d'Electronique et des Systèmes — *Oct 2017 - Nov 2020 (expected)*
  - Nonlinear acoustic microscopy: developed a new method of Nondestructive Testing (NDT) and ultrasonic imagery, using nonlinear effects of acoustic waves propagation.
  - Built and programmed a fully automated scanning system designed for acquisition, display, storage and computation of data associated with measurement and mapping of acoustic fields using LabVIEW.
  - Written ultrasonic field simulation programs in Python/Matlab

  R and Shiny developer — Montpellier
  University of Montpellier, Service of scientific culture — *Sep 2018 - Aug 2019*
  - Developed and integrated (in collaboration) a scientific software in R, that performed statistical analysis on stuffed animals regarding their morphology and emotional effects on their owners.
  - More than 850 contributions from around the world composed the database of the analysis.

  LabVIEW and signal processing lecturer (MCE) — Montpellier
  University of Montpellier, EEA department — *Sep 2018 - Aug 2019*
  - Teached signal processing and programming/interfacing HW using LabVIEW.
- **Matlab developer/ Ultrasonic microscopy - Internship** — Montpellier
  Institut d'Electronique et des Systèmes — *Apr 2017 - Aug 2017*
  - Developed a simulation software using Matlab and confirmed the results on a scanning acoustic microscope.
- **Industrial IT - Internship** — Taroudant, Morocco
  AUTEC - Automation Technology — *June 2016 - Aug 2016*
  - Built (in collaboration) a supervising and archiving system (HW/SW), for data coming from SIEMENS PLCs of the agro-food factory COPAG, using: Python - Windev - RaspberryPI - SQL - PHP - TIA Portal.
- **3D modeled (in collaboration) an automatic CNC tool changer - Internship** — Tanger, Morocco
  CIS MED — *June 2015 - Aug 2015*

Education

- **Engineering degree in Embedded systems and industrial IT** — Fez, Morocco
  National School of Applied Sciences — *2012 - 2017*

Honors and awards

- **1st Place: ProtoTop competition** *Cité d'innovation, USMBA* — *Mai 2016*
- **1st Place: Google Startup Weekend competition** *Google for Entrepreneurs* — *Feb 2016*
- **Scholarship for excellence** *MT2E Association* — *2012 - 2017*

Technical skills and languages

- **French:** Proficient - **English:** Proficient - **German:** Intermediate - **Chinese:** Basic - **Arabic :** Native
- **Programming:** C/C++, ASM, Python, Matlab, LabVIEW, Windev, VHDL, Java, HTML, CSS, JS.
- **Hardware:** Embedded systems, Sensors implementation, Signal Acquisition, Fast Prototyping, 3D printing.
- **Protocols and OSs:** CAN, UART, I^2C, SPI, TCP/IP, (T)FTP, SSH, Linux, MacOS, Windows.
- **Version control:** Git. • **Typesetting:** LaTeX.
- **3D modeling and printing :** CATIA and Fusion 360. 3D printers setup and modding.

Personal projets

- **Machine learning :** Implemented a genetic algorithm on Processing (Java based) that converges to a given solution from an initial random population by evolving over generations.
- **IoT Weather station - Wifi provisioning - ESP8266 :** Built a wifi-connected weather station and made firmware allowing Wi-Fi provisionnin for an easy setup.
- **CAN protocol and embedded C:** Built and programmed a CAN communication system between Easypic7 and Armadeus embedded boards allowing the transfer/reception of commands and data from ultrasonic sensors.
- **ADAS simulations:** Made a simulation of ABS and ESP systems on Matlab Simulink.
- **IoT IR gesture control:** Built an IR based gesture recognition device. Integrated it in different systems/uses.

Interests

- Artificial Intelligence • Maker/DIY • Societé Française d'Acoustique • Rotaract Club • Cycling • Swimming